

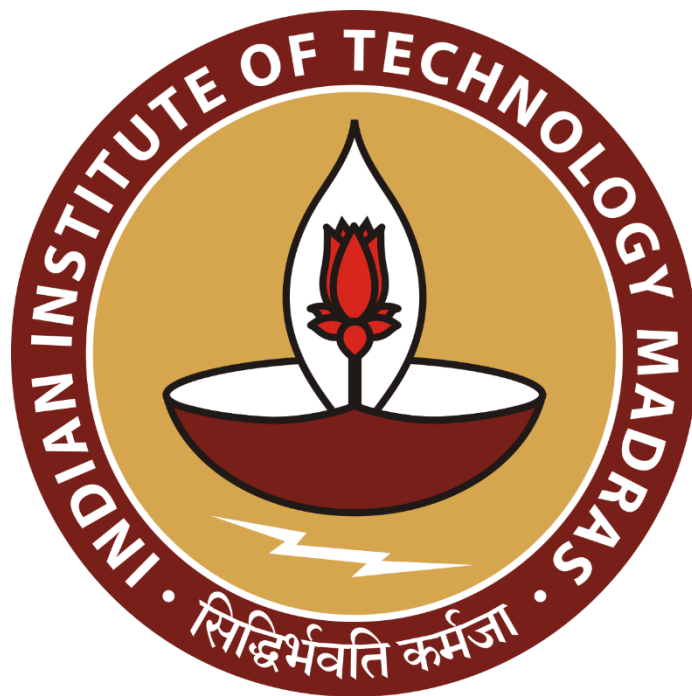
Daily sales analysis of a cafe to increase efficiency and profits

A Mid-Term report for the BDM capstone Project

Submitted by

Name: Vedaant Arora

Roll number: 24ds3000056



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai Tamil Nadu, India, 600036

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1 Executive Summary

This project studies the daily business performance of Coffee Beans, a small cafe in Anand Vihar, Delhi. The cafe owner keeps a handwritten record of sales and expenses, but the data has never been analysed before. Because of that, the cafe struggles with two main problems. First, sales change a lot from day to day which makes it difficult to plan inventory or predict weekly and monthly income. Second, although expenses are tracked, there is no clear understanding of where unnecessary spending is happening or which days cause losses.

I collected 150 days of data directly from the cafe and digitized it into Excel. After removing the days where both sales and expenses were zero or missing, there were 133 operating days that were suitable for analysis. The work was done using Excel and Python in Google Colab for cleaning, summarizing and plotting the data.

The descriptive statistics show that the cafe actually performs well most of the time. On more than 80 percent of the operating days, there is a positive profit. A typical day earns around ₹690 in profit. However, a few days with very high expenses for things like salaries, rent or bulk stock purchases pull the overall average daily profit down to just ₹40. In other words, most days are good but a few very bad days ruin the financial picture.

To investigate this, I will be using simple time trend analysis to understand demand patterns and cost ratio checks to find the days with poor expense control. The goal is to help the cafe owner make better decisions on stocking and spending so that profits become more stable in the future.

2 Proof of Originality

Business Name: Coffee Beans

Address: DDA market, Anand Vihar, Delhi-110092

Owner's Name: Mr. Swastik Khurana

Video of interaction with business owner: [Interaction Video](#)

Letter: [Letter](#)

Google Photos folder with photos of all sales data: [Photos](#)

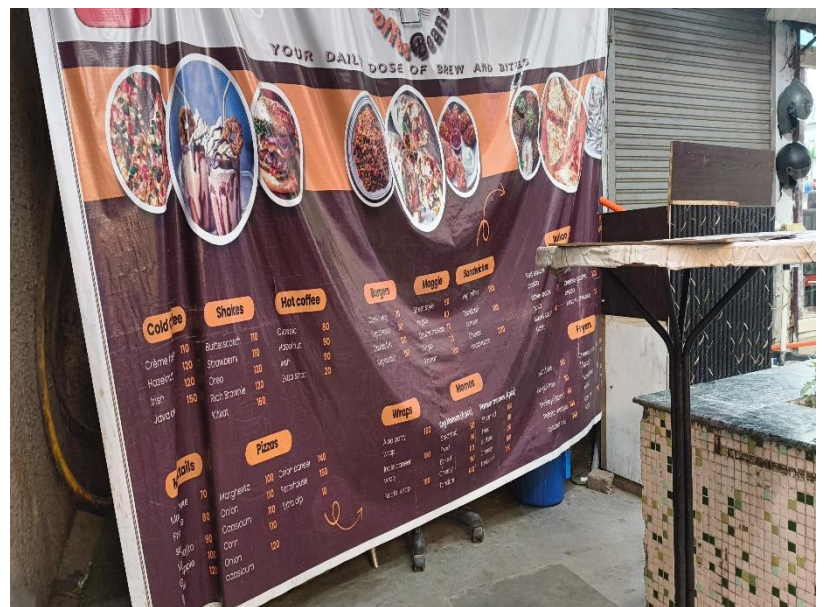


Fig 2.1: The cafe and it's seating.

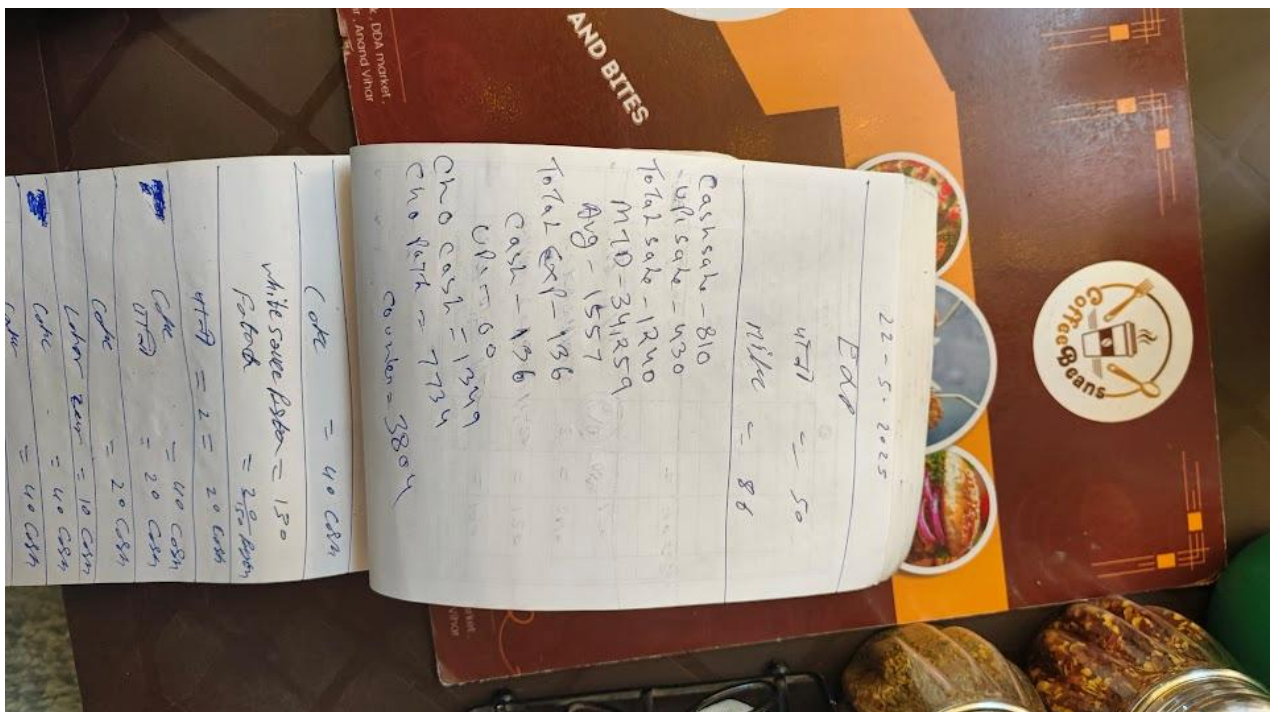


Fig 2.2: Photo of one of the pages of the record book (all the photos are attached in the google photos link present above)

3 Metadata

The cafe records all business information in a handwritten Record Book maintained by the owner. This book acts as the primary source of data for this project. I personally digitized the entries into Excel for analysis.

3.1 Information about the log book:

The cafe records the following data in the record book:

- Cash Sale: Sales done throughout the day using cash.
- UPI Sale: Sales done throughout the day using UPI.
- Total sales: Combined daily sales via cash and UPI.
- MTD: The number of sales throughout the month till the present day.
- Avg: Average sales of the month.
- Total expenditure: Total expenditure on the day (Further broken into cash and UPI expenditure).
- CHO cash and CHO Paytm: Tracks the amount remaining in each payment method.
- Counter: Money present at the counter.

This dataset contains daily-level data of Coffee Beans, a local cafe based in Anand Vihar, Delhi. The dataset was compiled from physical records and daily logs maintained by the cafe owner.

3.2 Dataset Format:

- File Format: .xlsx (Microsoft Excel)
- Total Rows: 150 daily entries
- Time Period: 1st February 2025 to 30th June 2025
- Currency: Indian Rupee (₹)
- Operating days (used for analysis): 133
- Excluded days: 17

3.3 Columns in the Dataset:

Column Name	Description
Date	Date of record (DD-MM-YYYY)
Day	Day of the week
Sales	Total sales done in the day
Expense	Total cost incurred on that day (raw materials, wages, etc.)
Profit	Calculated by me for a day to day understanding of profitability.
Remarks	Any notes (like holiday, Data unavailable, worker salary etc.)

Table 3.1 Dataset Description

3.4 Cleaning and Data Selection:

- A day is considered an Operating Day if at least one of Sales or Expense was recorded in the register.
- Days where both values were zero or missing were excluded because they represent cafe closure or absence of reliable data. This rule prevents distortion of averages and gives a realistic picture of operations.

3.5 Handling Missing Data:

- A number of dates in April and May had missing values due to lack of entries in the register
- If Sales or Expense alone was missing, the other value was still used (since the cafe can spend without earning or can earn using existing stock)
- Comments were used wherever available to help interpret spikes or losses.

4 Descriptive statistics

This section summarizes the café's financial performance across the 133 Operating Days where at least one of Sales or Expense was recorded. These Operating Days provide a reliable basis to understand how the café performs under normal business conditions.

The statistical measures presented here reflect the overall distribution of Sales, Expense and Profit, along with indicators of profitability and volatility.

4.1 Summary of Sales, Expense and Profit

Table 4.1 Stats of Sales, Expense and Profit

Descriptive statistics (operating days only)			
	Sales	Expense	Profit
count	133.00	133.00	133.00
mean	1490.73	1451.14	39.59
std	955.37	3648.19	3617.99
min	150.00	0.00	-23362.00
25%	910.00	141.00	190.00
50%	1290.00	370.00	690.00
75%	1820.00	916.00	1343.00
max	6200.00	27267.00	4609.00

Interpretation: Most days are profitable with a typical profit of around ₹690. However, a few days have extremely high expenses which result in large losses. This pulls the average daily profit close to zero, even though the median shows strong daily performance.

4.2 Profit Distribution

Table 4.2 Profit Distribution across days

Profit distribution across operating days			
	Category	Days	Share_%
0	Profit > 0	108	81.2
1	Profit < 0	25	18.8
2	Profit = 0	0	0.0

Interpretation: The café earns profit on more than four out of every five operating days. This shows that the business model is generally working well.

4.3 Cost Efficiency

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=== Expense ratio & Profit margin statistics ===

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	Expense_ratio	Profit_margin
count	133.0000	133.0000
mean	1.1284	-0.1284
std	2.8279	2.8279
min	0.0000	-18.6226
25%	0.1003	0.2884
50%	0.3107	0.6893
75%	0.7116	0.8997
max	19.6226	1.0000

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=== Volatility (Coefficient of Variation) ===

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Variable	Mean	StdDev	Coeff_of_Variation
0 Sales	1490.7293	955.3738	0.6409
1 Profit	39.5940	3617.9870	91.3772

Table 4.3 Expense Ratio and Profit margin stats

Table 4.4 Volatility of Sales and Profit

Interpretation: On a normal day, expenses are around 30 percent of revenue, leading to a high margin. But large expense events on a few days create negative average profit margin overall.

4.4 Insights from data

- Profitability is strong on most days but heavily damaged by occasional high-expense days
- Sales inconsistency explains the owner's concerns about income predictability
- Expense spikes are the primary reason for unstable profit
- Improving cost planning and spreading fixed expenses more evenly can significantly stabilize profits

5 Detailed Explanation of Analysis Process and Method

This section explains how I took the café's raw handwritten register data and turned it into something I could actually analyse. I used Excel and Python (Google Colab) to clean the data, calculate basic numbers and create visualizations. After filtering out days where nothing was recorded, I worked with 133 operating days of data.

5.1 Cleaning and Preparing the data

- Entered the handwritten register data into Excel
- Fixed the Date format and sorted records properly
- Checked the Sales, Expense and Profit columns for missing entries

- Kept a day if at least one of Sales or Expense was recorded
- Removed days where both were zero or not written at all
- Calculated Profit for each day (Sales minus Expense)

This helped remove days when the café was closed or the owner forgot to write anything down.

5.2 Methods for Understanding Sales Instability

The owner wants to know why income is unpredictable. For this, I used:

- Basic descriptive statistics
- Day-of-week analysis to see patterns
- A 7-day moving average to smooth out daily noise

These simple methods are enough to spot trends like weekend peaks or mid-week drops, without using advanced forecasting yet.

5.3 Methods for Checking Expense Issues

Some days have big losses even when sales look fine. To see what is going wrong, I used:

- Expense Ratio: To check if spending is too high compared to sales on certain days.
- Profit Margin: To see how much of the revenue the café actually keeps
- Identifying outlier loss days: Spotting extremely high negative profits
- Using the Comment notes: To understand real reasons like salary or rent

This shows exactly when and why profits collapse.

5.4 Tools used

- Excel: Recording, checking and storing values.
- Python (Pandas, Numpy, Matplotlib): Cleaning, Calculations, Tables and Charts

5.5 Why these Methods

Since the café's data is recorded as daily totals, the analysis first focuses on understanding overall financial patterns rather than item-level behaviour. The selected descriptive methods help measure

revenue variability, identify loss-driving expense events and reveal when profitability becomes unstable.

These techniques are appropriate because they directly address the café's two main issues:

- (1) unpredictable income, and
- (2) unexpected financial losses on certain days.

They also create a clear base for applying forecasting and cost optimisation methods in the final phase of the project.

6 Results and Findings

This section presents the key findings from the analysis performed on 133 operating days. Each graph is followed by a brief interpretation focused on the café's business concerns.

6.1 Sales Trend and Moving Average (7-day)

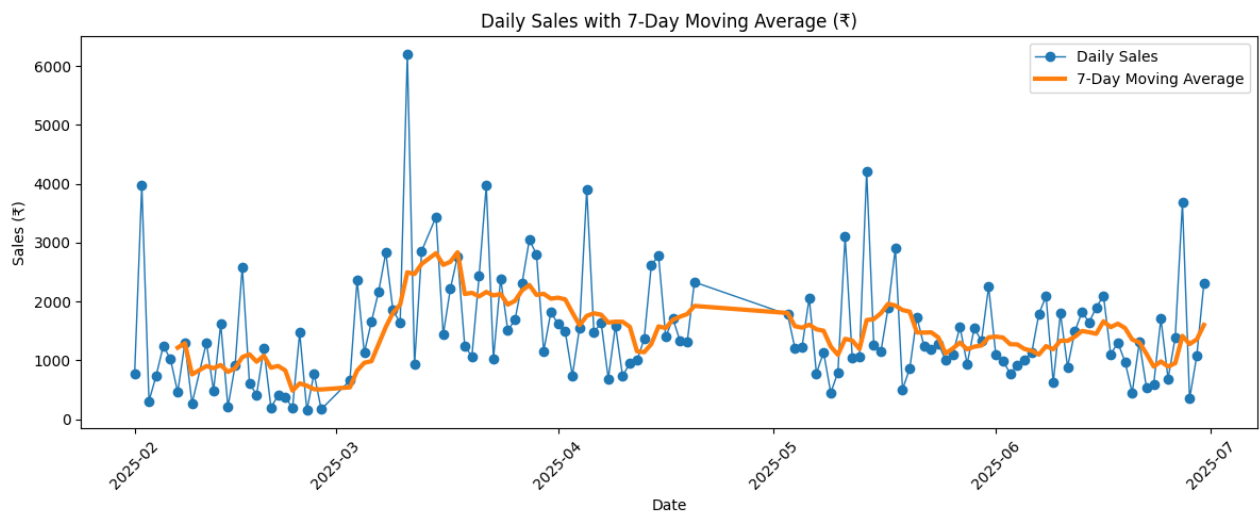


Figure 6.1 Chart of 7 days moving average

Interpretation: Sales change a lot from day to day, but the moving average shows a regular increase toward weekends followed by drops on Mondays and Thursdays. This confirms that customer footfall is not consistent and supports the store owner's concern about income unpredictability.

6.2 Weekday Analysis of Sales and Profit

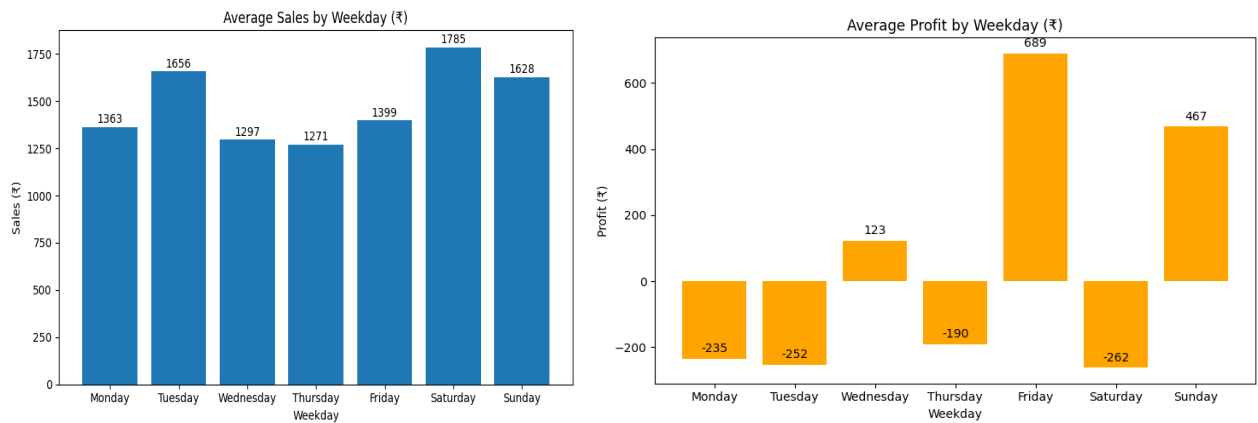


Figure 6.2 Average Sales and Profit

Interpretation: Saturday and Tuesday have high sales but low average profit, which means higher spending on those days reduces efficiency. Friday shows strong profits even with moderate sales, indicating better cost control that day.

6.3 Expense to Sales Comparison



Figure 6.3 Expense vs sales chart

Interpretation: Expense spikes appear on very specific dates and are mostly linked to salary or rent payments (based on comments). These events cause major losses and are the biggest reason for unstable profit.

6.4 Month-wise Total Profit

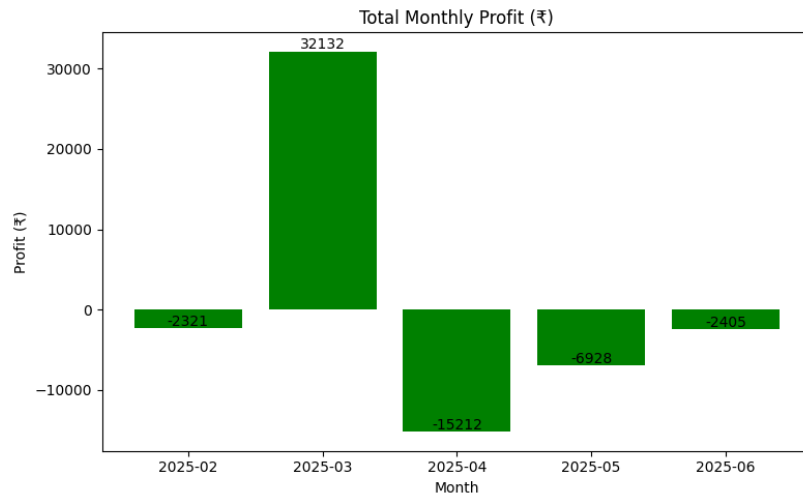


Figure 6.4 Monthly Profit

Interpretation: March shows strong overall profitability. April, May and June show losses because multiple expense spikes fall in those months. Better scheduling of expense events could improve profit stability.

The café performs positively on most operating days, with nearly 81 percent of days showing a profit. A typical day earns close to ₹690, which indicates that the core business model works well. However, the overall average profit drops sharply to around ₹40 because a few days include very large expenses related to salaries, rent or bulk purchases. These loss-heavy days have a major impact on monthly profit even though the business is generally healthy day-to-day.

Sales also show a weekday pattern. Weekends and early-week days such as Saturday and Tuesday see higher sales, but profit does not always follow, since costs also rise on those days. In contrast, Fridays have relatively moderate sales but strong profits due to better cost control. Monthly totals reflect these issues too. March has a clear positive outcome, but April to June show losses because several major expenses fall together in those months.

Overall, the café does not face a demand problem. Its main challenge is unstable cost planning. By distributing major expenses more evenly and managing spending better on high-sales days, the café can turn consistent daily success into reliable monthly profit.